Ch0 Theory of Interest

Accumulation Function Simple interest: $a(t) = (1 + rt)$, Compound interest: $a(t) = (1 + r)^t$. $r^{(p)}$ = Nominal yield compounded p times a year, interest in one period = $\frac{r^{(p)}}{p}$, interest per year = $(1 + \frac{r^{(p)}}{p})^p$

Effective Interest On the interval $[t_1, t_2] = \frac{a(t_2)-a(t_1)}{a(t_1)}$

Continuously Compounded Rate $a(t) = e^{rt}$. $e^r > (1+\frac{r}{p})^p \geq 1+r$ for any $r > 0$ and $p \geq 1$

Equivalent Interest Rate Two rates are equivalent iff they yield the same accumulation amount over a year, $(1 + \frac{r^{(p)}}{p})^p = (1 + \frac{r^{(q)}}{q})^q$

Effective annual interest rate $1 + r_e = (1 + \frac{r^{(p)}}{p})^p$. For any positive integer p, $r_e \equiv (1 + \frac{r^{(p)}}{p})^p - 1 \geq r^{(p)}$, $a(t) = (1 + r_e)^t$

Present Value A.k.a discounting. For $C = \{(c_1, t_1), \ldots, (c_n, t_n)\}$, $PV(C) = \sum_{i=1}^n \frac{c_i}{a(t_i)}$. Notation: $C = \{(c_1, 0) \ldots, (c_n, n-1)\} = (c_1, c_2, \ldots, c_n)$

Principle of Equivalence When IR and $a(t)$ are the same, two cash flows are equivalent iff they have the same present value.

Time Value $TV_t C = PV(C) \times a(t)$, $PV(C) = \frac{TV_t(C)}{a(t)}$. For $0 < t_1 < t_2$, $TV_{t_2}(C) = \frac{a(t_2)}{a(t_1)} \times TV_{t_1}(C)$

Equation of Value For a cash flow $C = \{(c_1, t_1), (c_2, t_2), \ldots, (c_n, t_n)\}$, $PV(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i}} = 0$

Law of One Price PV(what you pay) = PV(what you receive)

Newton Raphson Method To estimate non-negative root r of the equation of value $PV(C) = 0$, a.k.a yield/**I**nternal **R**ate of **R**eturn

- First use Intermediate Value Theoreom to pin down a small interval containing the root
- Then pick an arbitrary point as the starting point α_0
- $\alpha_1 = \alpha_0 - \frac{f(\alpha_0)}{f'(\alpha_0)}$, $\alpha_{n+1} = \alpha_n - \frac{f(\alpha_n)}{f'(\alpha_n)}$

Example Find the IRR of $C = (-2, 1, 1, 1)$ to 4 dp. Construct $PV(C) = \cdots = 0$. Let $x = 1 + r$, set $f(x) = 0$, find $f'(x)$. Test values of $f(x)$ at $x = 1, 1.1, 1.2 \ldots$, observing where it crosses the y-axis. Use IVT to establish a range and set α_0 within the range.

Ch1 Financial Instruments & Term Structure

Certificate of Deposit (CD) Time deposit with a bank that cannot be drawn on demand (penalty for withdrawing early). Value after t years = $P \times a(t)$, earnings = $P \times (a(t) - 1)$. Interest rate earned for the kth year = $P \times (a(k) - a(k-1))$ (to calculate penalty)

Stock Ownership of a company. Example: A share is expected to pay \$10 dividends at the end of the year, which are expected to grow 3% per year. Assume market return is 15%. Calculate the expected stock price. ($a(t) = 1.15^t$, $PV(C) = \sum_{t=1}^\infty \frac{10 \times (1+3\%)^{t-1}}{a(t)}$)

Treasury bills (T-bills) Short term (within one year) debt issued by the government. Issued at discount to par, have no coupon rate, mature at par value. $P(1 + \lambda \times M) = R = F$ where M is years till maturity. Yield $\lambda = \frac{(F/P)-1}{M}$

Commercial Paper Short term (1-2 months) unsecured debt note by a company. Issued at discount to par, no coupon payment, mature at par. Return is difference between maturity value and purchase price. Earnings = $P_1 - P_0$, rate of return = $\frac{P_1-P_0}{P_0}$, nominal rate of return = $\frac{P_1-P_0}{P_0} \times \frac{365}{t_2-t_1}$

Bond Valuation By default, yields are quoted on an **annual** basis.

$$P = \frac{R}{(1 + \frac{\lambda}{m})^n} + \sum_{i=1}^n \frac{cF/m}{(1 + \frac{\lambda}{m})^i}$$

P = current price, F = face value (used by coupon rate), R = redemption value at maturity, c = nominal coupon rate, m = no. of coupon payments per year, n = total no. of coupon payments, λ = nominal yield

Bond yield is the IRR of the bond at its current price.

Assume $R = F$ unless otherwise stated. When $R = F$,

$$P = F\left[\frac{1}{1 + (\frac{\lambda}{m})^n} + \frac{c}{\lambda}\left(1 - \frac{1}{(1 + \frac{\lambda}{m})^n}\right)\right] = F + F\left(\frac{c-\lambda}{\lambda}\right)\left[1 - \frac{1}{(1 + \frac{\lambda}{m})^n}\right]$$

(i)$P > F$ iff $c > \lambda$ (premium), (ii)$P = F$ iff $c = \lambda$ (at par), (iii)$P < F$ iff $c < \lambda$ (at discount)

Example A 30-year bond has face value 100 and pays coupons semi-annually. The nominal coupon rate is 4% for the first 10 years, 6% for the next 10 years, and 8% for the last 10 years. The price and redemption value are P and $P + 30$ respectively. If the nominal yield is 7%, find P. ($P = \sum_{i=1}^{20} \frac{2}{(1+3.5\%)^i} + \sum_{i=1}^{20} \frac{3}{(1+3.5\%)^{i+20}} + \sum_{i=1}^{20} \frac{4}{(1+3.5\%)^{i+40}} + \frac{P+30}{(1+3.5\%)^{60}}$. Note that $a(t)$ is 1.035^t for half year period.)

Price after kth payment $P_{k+1} = P_k(1 + \frac{\lambda}{m}) - cF/m$

Zero Coupon Bond Price $P = \frac{R}{(1+\lambda)^n}$

Perpetual Bond Bond that never matures ($n \to \infty$) $P = \frac{cF}{\lambda}$

Taylor Series Approximate value of f near values of a

$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^n(a)}{n!}(x-a)^n + \ldots$

Price resulting from change in yield

$P(\lambda + \Delta\lambda) = P(\lambda) + P'(\lambda) \cdot \Delta\lambda + \frac{P''(\lambda)}{2} \cdot (\Delta\lambda)^2 + \cdots + \frac{P^n(\lambda)}{n!} \cdot (\Delta\lambda)^n + \ldots$

Percentage change of bond price

$\frac{P(\lambda+\Delta\Lambda)-P(\lambda)}{P(\lambda)} = \frac{P'(\lambda)}{P(\lambda)} \cdot \Delta\lambda + \frac{1}{2}\frac{P''(\lambda)}{P(\lambda)} \cdot (\Delta\lambda)^2 + \cdots + \frac{1}{n!}\frac{P^n(\lambda)}{P(\lambda)} \cdot (\Delta\lambda)^2 + \ldots$

Modified Duration $D_M = -\frac{P'(\lambda)}{P(\lambda)}$

Price Sensitivity Formula $\frac{P(\lambda+\Delta\Lambda)-P(\lambda)}{P(\lambda)} \approx -D_M(\lambda) \cdot \Delta\lambda$

On the price-yield curve, $-D_M P$ is the slope of the tangent line at λ

Estimation of Price Change

New price $P(\lambda + \Delta\lambda) \approx P(\lambda) - (D_M P) \cdot \Delta\lambda$

Change in bond price $\Delta P = P(\lambda + \Delta\lambda) - P(\lambda) \approx -(D_M P) \cdot \Delta\lambda$

Example Price of bond is \$110, yield is 7%, modified duration is 5. Estimate the price of the bond if its yield falls to 6%.

Note: modified duration depends on the type of yield.

Notation: For $C = \{(c_i, t_i) : i = 1, 2, \ldots, n\}$, $tC = \{(t_i c_i, t_i : i = 1, 2, \ldots, n\}$

For a bond with cash flow $C = \{(cF/M, i/m) : i = 1, 2, \ldots, n\} \cup \{(R, n/m)\}$, suppose λ is nominal yield compounded k times a year.

$$P(\lambda) = \frac{R}{(1 + \lambda/k)^{kn/m}} + \sum_{i=1}^n \frac{cF/M}{(1 + \lambda/k)^{ki/m}} = PV(C)$$

$$P'(\lambda) = -\frac{\frac{nR}{m}}{(1 + \frac{\lambda}{k})^{\frac{kn}{m}+1}} - \sum_{i=1}^n \frac{(\frac{i}{m})(\frac{cF}{m})}{(1 + \frac{\lambda}{k})^{\frac{ki}{m}+1}}$$

$$= \left(-\frac{1}{1 + \frac{\lambda}{k}}\right)\left[\frac{n}{m} \cdot \frac{R}{(1 + \frac{\lambda}{k})^{\frac{kn}{m}}} + \sum_{i=1}^n \frac{i}{m} \cdot \frac{\frac{cF}{m}}{(1 + \frac{\lambda}{k})^{\frac{ki}{m}}}\right]$$

$$= \left(-\frac{1}{1 + \frac{\lambda}{k}}\right) \cdot PV(tC)$$

$\Rightarrow D_M(\lambda) = \frac{P'(\lambda)}{P(\lambda)} = \frac{1}{1+\frac{\lambda}{k}} \cdot \frac{PV(tC)}{PV(C)}$. When $k = 1$, $D_M(\lambda) = \frac{1}{1+\lambda} \cdot \frac{PV(tC)}{PV(C)}$ (i.e. λ is effective annual yield)

If λ is continuously compounded yield, $a(t) = \exp(\lambda t)$.

$$P(\lambda) = \frac{R}{\exp(\lambda(n/m))} + \sum_{i=1}^n \frac{cF/m}{exp(\lambda(i/m))} = PV(C)$$

$$P'(\lambda) = -\frac{nR/M}{\exp(\lambda(n/m))} - \sum_{i=1}^n \frac{cF/m}{\exp(\lambda(i/m))}$$

$$= -\left[\frac{n}{m} \cdot \frac{R}{\exp(\lambda(n/m))} + \sum_{i=1}^n \frac{i}{m} \cdot \frac{cF/m}{\exp(\lambda(i/m))}\right] = -PV(tC)$$

$\Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{PV(tC)}{PV(C)}$

Macaulay Duration (duration for short): $D(\lambda)$ is the modified duration w.r.t the bond's continuously compounded yield.

$MacD = D = \frac{PV(tC)}{PV(C)}$

If λ is nominal yield compounded k times per year, $D_M(\lambda) =$

$\frac{1}{1+\frac{\lambda}{k}}D(\lambda)$.
If λ is effective annual yield, $D_M(\lambda) = \frac{1}{1+\lambda}D(\lambda)$.
If λ is continuously compounded yield, $D_M(\lambda) = D(\lambda)$.
Note: Macaulay Duration does not depend on the type of yield.
Example A 5-year bond pays annual coupons of 5% and is priced at par. Calculate the modified duration and duration w.r.t (i) the effective yield, (ii) continuously compounded yield, and (iii) semiannually compounded yield. (Find $P(\lambda)$ and $P'(\lambda)$ where λ is **nominal yield**, or find $PV(tC)$ and $PV(C)$ using **effective annual yield**.)
Macaulay Duration for Bond $P(\lambda) = F + F(\frac{c-\lambda}{\lambda})\left[1-(1+\frac{\lambda}{m})^{-n}\right]$,

$$P'(\lambda) = F(-\frac{c}{\lambda^2})\left[1-(1+\frac{\lambda}{m})^{-n}\right] + F(\frac{c-\lambda}{\lambda})\left[\frac{n}{m}(1+\frac{\lambda}{m})^{-n-1}\right]$$

$$-D_M = \frac{P'(\lambda)}{P(\lambda)} = \ldots = -\frac{1}{\lambda} + \frac{\frac{1}{c}\left[1+\frac{c-\lambda}{m}\cdot n\cdot\frac{1}{1+\lambda/m}\right]}{(1+\frac{\lambda}{m})^n - 1 + \lambda/c}$$

$$D(\lambda) = (1+\frac{\lambda}{m})D_M = \frac{1+\mu}{m\mu} - \frac{1+\mu+n\cdot(\gamma-\mu)}{m\gamma[(1+\mu)^n-1]+m\mu}$$

(Let $\mu = \frac{\lambda}{m}$, $\gamma = \frac{c}{m}$) As $n\to\infty$, for a perpetual bond, $D = \frac{1+\mu}{m\mu} = \frac{1+\lambda/m}{\lambda}$. When bond is priced at par, $\mu = \gamma$, $D = \frac{1+\mu}{m\mu}\left(1-\frac{1}{(1+\mu)^n}\right)$
Generalization to Other Cash Flows For $C = \{(c_i, t_i) : i = 1, 2, \ldots, n\}$. Let $P(\lambda) = PV(C)$. $D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)}$.

$$P(\lambda) = \sum_{i=1}^n \frac{c_i}{a(t_i)} = \sum_{i=1}^n \frac{c_i}{(1+\frac{\lambda}{k})^{kt_i}} = PV(C)$$

$$P'(\lambda) = -\sum_{i=1}^n \frac{c_i t_i}{(1+\frac{\lambda}{k})^{kt_i+1}} = -\frac{1}{1+\frac{\lambda}{k}}\sum_{i=1}^n \frac{c_i t_i}{(1+\frac{\lambda}{k})^{kt_i}}$$

$$= -\frac{1}{1+\frac{\lambda}{k}}\sum_{i=1}^n \frac{c_i t_i}{a(t_i)} = -\frac{1}{1+\frac{\lambda}{k}}\cdot PV(tC)$$

$$\Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{1}{1+\frac{\lambda}{k}}\cdot\frac{PV(tC)}{PV(C)}$$

If $k = 1$ i.e. λ is effective annual interest rate, $D_M(\lambda) = \frac{1}{1+\lambda}\cdot\frac{PV(tC)}{PV(C)}$
If λ is continuously compounded interest rate, $a(t) = \exp(\lambda t)$

$$P(\lambda) = \sum_{i=1}^n \frac{c_i}{a(t_i)} = \sum_{i=1}^n \frac{c_i}{\exp(\lambda t_i)} = PV(C)$$

$$P'(\lambda) = -\sum_{i=1}^n \frac{c_i t_i}{\exp(\lambda t_i)} = -\sum_{i=1}^n \frac{c_i t_i}{a(t_i)} = -PV(C)$$

$\Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{PV(tC)}{PV(C)} \equiv D(\lambda)$
If λ is nominal yield compounded k times a year, $D_M(\lambda) = \frac{1}{1+\frac{\lambda}{k}}D(\lambda)$
Example The duration of $C = (0, 6, 0, 2)$ at an effective annual interest rate r is $\frac{85}{61}$. Find r and the modified duration. ($\frac{85}{61} = D = \frac{PC(tC)}{PV(C)}$. Substitute $x = \frac{1}{1+r}$, solve for x and r. $D_M = \frac{1}{1+r}\cdot D$)
Macaulay Duration (Time Weighted Average)
For $C = \{(c_i, t_i) : i = 1, 2, \ldots, n\}$, $D = \frac{PV(tC)}{PV(C)} = \frac{\sum_{i=1}^n t_i PV(c_i)}{\sum_{i=1}^n PV(C_i)} = \sum_{i=1}^n w_i t_i$, where $w_i = \frac{PV(c_i)}{\sum_{i=1}^n PV(c_i)}$
If $c_i \geq 0$ for all i, then $t_1 \leq D \leq t_n$. For a zero-coupon bond, $D = T$. The definition of D can be extended to any infinite cash flow.
Convexity $C(\lambda) = \frac{P''(\lambda)}{P(\lambda)}$. 2nd order approximation of change in bond price for a change in yield. Consider the quadratic term in the Taylor series: $P(\lambda+\Delta\lambda) \approx P(\lambda) + P'(\lambda)\Delta\lambda + \frac{P''(\lambda)}{2}(\Delta\lambda)^2$
Percentage change in bond price
$\frac{P(\lambda+\Delta\lambda)-P(\lambda)}{P(\lambda)} \approx \frac{P'(\lambda)}{P(\lambda)}\Delta\lambda + \frac{1}{2}\frac{P''(\lambda)}{P(\lambda)})(\Delta\lambda)^2 = -D_M\Delta\lambda + \frac{1}{2}C(\lambda)^2$.
$\Delta P = P(\lambda+\Delta\lambda) - P(\lambda) \approx P[-D_M\Delta\lambda + \frac{1}{2}C(\Delta\lambda)^2]$
Example The modified duration and convexity of a bond are 10.62 and 182.92 respectively. The annual YTM changes from 9 to 11% per annum. Find an approximation for the percentage change in bond price. ($D_M = 10.62$, $C = 182.92$, $\Delta\lambda = 0.02$, find $\frac{\Delta P}{P}$)

Note: C depends on the type of interest rate.
Let $C = \{(c_i, t_i) : i = 1, 2, \ldots, n\}$. Suppose k is nominal IR compounded k times a year.

$$P(\lambda) = \sum_{i=1}^n \frac{c_i}{(1+\lambda/k)^{kt_i}} = PV(C)$$

$$P'(\lambda) = -\sum_{i=1}^n \frac{c_i t_i}{(1+\lambda/k)^{kt_i+1}}$$

$$P''(\lambda) = \sum_{i=1}^n \frac{c_i t_i(t_i+1/k)}{(1+\lambda/k)^{kt_i+2}} = \sum_{i=1}^n \frac{c_i t_i^2}{(1+\lambda/k)^{kt_i+2}} + \sum_{i=1}^n \frac{c_i t_i}{(1+\lambda/k)^{kt_i+2}}$$

$$= \frac{1}{(1+\lambda/k)^2}(PV(t^2C) + PV(tC)/k)$$

$$C(\lambda) = \frac{P''(\lambda)}{P(\lambda)} = \frac{1}{(1+\lambda/k)^2}\cdot\left(\frac{PV(t^2C)}{PV(C)} + \frac{1}{k}\cdot\frac{PV(tC)}{PV(C)}\right)$$

When $k = 1$ i.e. λ is effective annual IR, then $C(\lambda) = \frac{1}{(1+\lambda)^2}\cdot(\frac{PV(t^2C)+PV(tC)}{PV(C)})$
If λ is continuously compounded, $P(\lambda) = \sum_{i=1}^n c_i\exp(-t_i\lambda) = PV(C)$, $P'(\lambda) = -\sum_{i=1}^n c_i t_i\exp(-t_i\lambda) = -PV(tC)$, $P''(\lambda) = \sum_{i=1}^n c_i t_i^2\exp(-t_i\lambda) = PV(t^2C)$. $\Rightarrow C(\lambda) = \frac{PV(t^2C)}{PV(C)}$
Macaulay Convexity $MacC = \frac{PV(t^2C)}{PV(C)}$. Macaulay convexity does not depend on the type of IR. It is the weighted sum of time squared.
If λ is nominal IR compounded k times a year, $C(\lambda) = \frac{1}{(1+\lambda/k)^2}(MacC(\lambda) + D(\lambda)/k)$.
If λ is effective annual IR, $C(\lambda) = \frac{1}{(1+\lambda)^2}(MacC(\lambda) + D(\lambda))$.
If λ is continuously compounded IR, $C(\lambda) = MacC(\lambda)$.
Spot Rate Annual interest rate starting today ($t = 0$) until some time in the future, effectively the YTM of a ZCB maturing at t

$$(1+s_t)^t = (1+s_j)^j(1+f_{j,t})^{t-j}$$

$$(1+s_n)^n = (1+s_1)(1+f_{1,2})(1+f_{2,3})\ldots(1+f_{n-1,n})$$

Spot rate curve = spot rate against YTM
$P = \frac{F}{(1+s_n)^n} \Rightarrow s_n = (\frac{F}{P})^{1/n} - 1$
If λ is not known, use spot rate to find bond price.
Forward Rate Interest rate starting at $t_1 > 0$ lasting until $t_2 > t_1$, denoted f_{t_1,t_2}. Note: $f_{0,t} = s_t$. (Remember to subtract 1 from effective rate to get spot/forward rate)
Bootstrap Method Suppose $s_1, \ldots, s_{n-1}$ have already been determined. To estimate s_n, consider an n-period coupon paying bond with cash flow $(-P_n, cF, cF, \ldots, cF+F)$. Then $P_n = \frac{cF}{(1+s_1)} + \frac{cF}{(1+s_2)^2} + \cdots + \frac{cF}{(1+s_{n-1})^{n-1}} + \frac{cF+F}{(1+s_n)^n}$, so s_n can be determined.